MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/119-RG174

| THIS CABLE USES PVC MATERIAL AND IS NOT TO BE USED IN AEROSPACE APPLICATIONS. |

| NOTE: THE AIR FORCE HAS RESTRICTED THE USE OF PVC IN AEROSPACE AND GROUND SUPPORT APPLICATIONS. CABLES WITH PVC JACKETING SHALL BE USED FOR RETROFIT PURPOSES ONLY UNTIL AN ALTERNATE JACKET IS APPROVED. |

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-17.

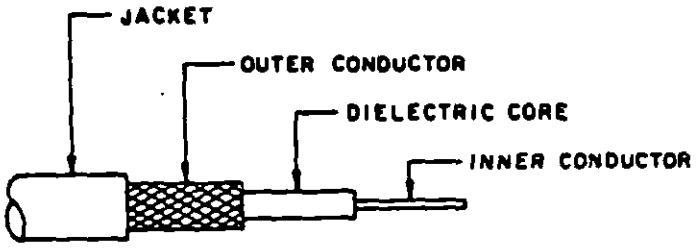


FIGURE 1. Configuration.

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TABLE I. Description.

Component	Construction details
Inner conductor	Seven strands of copper-covered steel wire, each strand .0063 inch diameter. Overall diameter: 0.0189 each ±0.0010.
Dielectric core	Type A-1: Solid polyethylene. Diameter: 0.060 inch ±0.003.
Outer conductor	Single braid of AWG No. 38 tinned copper wire. Diameter: 0.088 inch maximum. Coverage : 85.6% nominal. Carriers : 16 Ends : 4 Picks/inch : 16.3 ±10%
Jacket	Type IIa: PVC. Diameter: 0.110 inch ±0.005.

ENGINEERING INFORMATION:

Continuous working voltage: 1,100 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 9.67 ohms per 100 feet.

Elongation: 1 percent, minimum.

Tensile strength: 110 $klb_f/inch^2$, minimum.

Engineering note: This cable useful in general purpose, medium low temperature applications. (See connector series "SMA" and "SMB" per MIL-C-39012.)

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

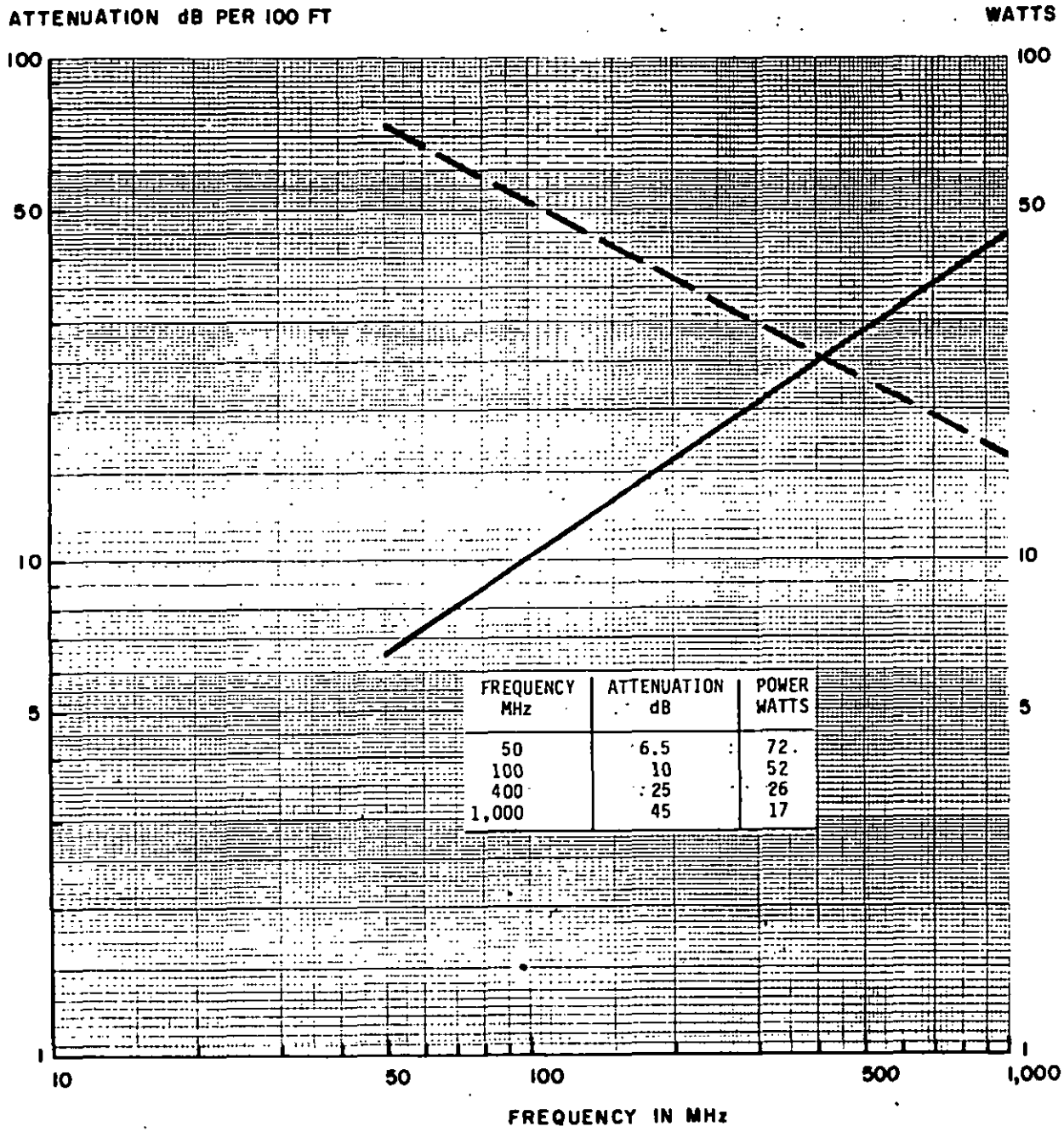
Adhesion of conductors:

Inner conductor to core: 4 pounds, minimum; 12 pounds, maximum.

Aging stability: +98°C ±2°C.

Stress crack resistance: Not applicable.

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FREQUENCY MHz	ATTENUATION dB	POWER WATTS
50	6.5	72.
100	10	52
400	25	26
1,000	45	17

Tabulated values are for reference only.
The values on the chart represent the requirements.

FIGURE 2. Power rating and attenuation.

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Outer conductor integrity: Not applicable.

(G) Cold bend: -55°C to ±2°C.

Dimensional stability: +85° ±2°C.

Inner conductor from core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 0.0095 pound per foot, maximum.

Electrical:

Continuity: Applicable.

(G) Spark test: 2,000 V rms, +10%, -0%.

(G) Voltage withstanding: 2,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,500 V rms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

(G) Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: M17/119-RG174.

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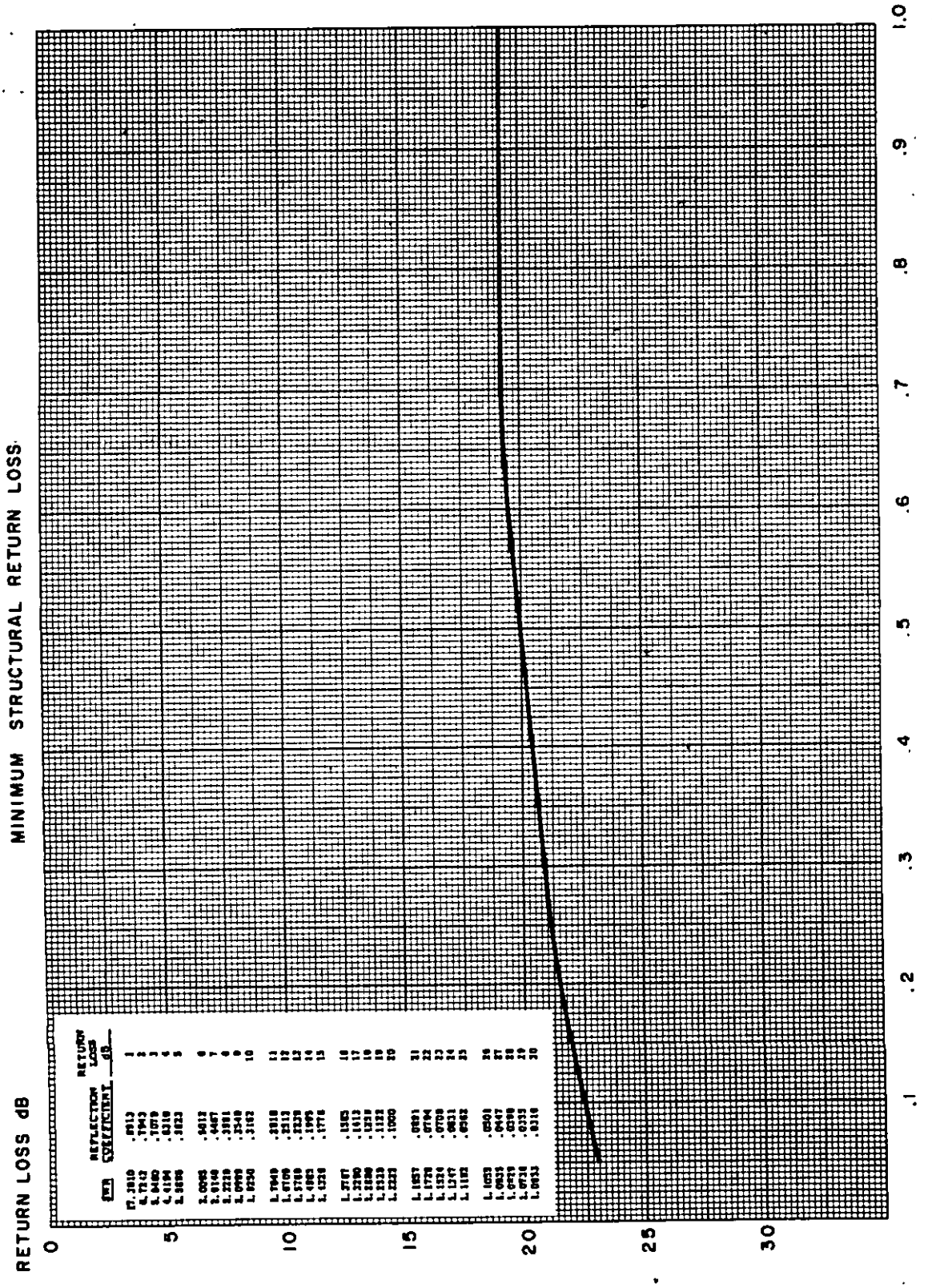


FIGURE 3. Structural return loss.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review Activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA- ES, IS

User Activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing Activity:
Army-CR

(Project 6145-0911-25)

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